

Muhammad Badar-Uz-Zaman

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PROFESSIONAL SUMMARY

Metallurgical Engineer with experience in foundry operations, materials science, and casting quality control. Skilled in alloy control, defect analysis, inspection, and compliance with ASME/ASTM standards. Focused on process improvement, quality assurance, and continuous improvement in ferrous and non-ferrous foundry environments.

WORK EXPERIENCE

Production and Quality Control Engineer | Madina Autos Industries PVT. Ltd 2025 – Present

- Managed in-process and final quality inspections of cast and aluminum components, ensuring compliance with defined specifications and manufacturing standards.
- Conducted spectroscopy testing to verify chemical composition of molten metal (CI & Aluminum HD2) and ensured material conformity before production release.
- Monitored Cupola Furnace melting, Green Sand Molding, and Molasses Core Making processes from a quality perspective to minimize defects and rejections.
- Performed defect identification and root cause analysis (porosity, shrinkage, sand defects, composition variation) and coordinated corrective actions with production teams.
- Maintained quality records, inspection reports, and process control data, supporting continuous improvement and consistent product quality.

Production and Quality Engineer | The Patriot Engineering Company PVT. Ltd 2025

- Hands-on exposure to green sand and resin sand molding, core making, melting in induction furnaces, and controlled pouring of gray and ductile iron castings.
- Monitored chemical composition, inoculation, and process parameters to achieve desired microstructure and mechanical properties.
- Conducted visual inspection, dimensional verification, and defect analysis for issues such as shrinkage, blowholes, cold shuts, and sand inclusions.
- Supported hardness testing, microstructure evaluation, and spectrometer analysis to ensure compliance with material specifications.
- Analyzed rejection trends and implemented corrective actions to improve yield and reduce scrap rates.

Internee – Inspection Engineer | Velosi Integrity & Safety Pakistan (Pvt.) 2024

- **API Inspection:** Acquired knowledge of API inspection techniques for pipelines, pressure vessels, storage tanks, and industrial equipment, focusing on safety and compliance requirements.
- **NDT Methods:** Gained hands-on exposure to Non-Destructive Testing (NDT) techniques such as Ultrasonic Testing (UT), Radiographic Testing (RT), Magnetic Particle Testing (MT), and Dye Penetrant Testing (PT) for weld and material evaluation.
- **Calibration & Lifting Equipment Inspection:** Assisted in calibration activities and inspection of cranes, hoists, slings, and other lifting tools/equipment to verify compliance with safety codes and operational standards.
- **Third-Party Inspection:** Supported independent and third-party inspections, observing how external verification ensures industry compliance, safety assurance, and quality validation.

- **Scrap Yard & Raw Material Handling:** Gained hands-on exposure to scrap yard operations, including sorting, grading, and preparation of raw materials for induction melting, ensuring proper material quality for steel production.
- **Induction Melting & Casting:** Observed and assisted in **induction furnace operations**, learning about charge preparation, melting practices, ladle handling, and temperature control. Acquired knowledge of **continuous casting processes**, including mold operations, billet formation, and cooling systems.
- **Rolling Mill Operations:** Supported production activities in the **rolling mill**, including reheating furnace monitoring, rolling passes, dimensional checks, and defect identification. Assisted in **quality control inspections** to ensure compliance with mechanical and dimensional standards.
- **Environmental Compliance & Emission Control:** Learned about **dust extraction systems, fume control units, and emission monitoring practices** implemented to meet environmental regulations and maintain workplace safety.

Internee Quality Control Engineer | Ravi Green Engineering PVT. Ltd

2024

- **Fabrication & Welding:** Gained hands-on exposure to fabrication methods and modern welding techniques, including process monitoring and weld quality evaluation.
- **WPS Development:** Learned the preparation, review, and application of **Welding Procedure Specifications (WPS)** in line with **ASME and ASTM standards**.
- **Inspection & NDT:** Assisted in **Non-Destructive Testing (NDT)**, visual inspections, and weld integrity checks to ensure compliance with engineering and safety requirements.
- **Quality Control & Drawings:** Strengthened skills in **technical drawing interpretation, dimensional verification, and material inspection**, supporting overall quality assurance processes.

EDUCATION

Bachelor of Science in Metallurgy and Materials Engineering

2021-2025

Institute of Metallurgy and Materials Engineering
Faculty of Chemical and Materials Engineering
University of the Punjab, Lahore

ADDITIONAL INFORMATION

Technical Skills

- **Metallurgy & Materials Science:** Phase diagrams, Heat treatment, Phase transformations, Corrosion resistance
- **Manufacturing & Processing:** Casting, welding, Forging, Rolling, Powder Metallurgy, Additive manufacturing (3D printing)
- **Materials Testing & Characterization:** Mechanical testing (tensile, hardness, impact, fatigue), Non-Destructive Testing (NDT), Scanning Electron Microscopy (SEM), X-ray Diffraction (XRD), Transmission Electron Microscopy (TEM) Optical Microscopy
- **Computational & Simulation Tools:** MATLAB, Finite Element Analysis (FEA), SolidWorks, AutoCAD
- **Corrosion & Failure Analysis:** Electrochemical testing, Failure analysis, Root cause analysis (RCA)
- **Advanced Materials:** Composite materials, Nanomaterials, Biomaterials, Smart materials
- **Quality Control & Standards:** ASTM, ISO, ASME standards.

Soft Skills:

- Presentation Skills
- MS Office
- Word, Excel and PowerPoint
- MATLAB
- Solid Cast
- Auto CAD
- Team Collaboration
- Safety Compliance & Risk Awareness
- Creativity & Innovation
- Goal-Oriented & Results-Driven
- Continuous Improvement & Process Innovation

AWARDS /ACTIVITIES

- Received the "Best Volunteer" Award for outstanding contributions in Two Day Workshop on Advance Aluminum Alloy Organized by Pakistan Advance Materials Forum (PAMF), SOAN Enterprise Islamabad and Institute of Metallurgy and Materials Engineering, PU, Lahore
- Work as a Volunteer to Presenting Institute of Metallurgy and Materials Engineering in 9th International Foundry Congress & Exhibition IFCE 2025
- Received "Best Poster Award" on Poster Competition Presentation at World Environment Day 22 March 2024 Organized by Institute of Metallurgy and Materials Engineering, IMME, PU.
- Received the Certificate of Participation as "Ambassador of METACOMP 2.0 | UET, LHR
- Participating on METACOMP 2.0 as a participant in Metallography/Quiz/Simulation/Poster Competition held on University of Engineering and Technology UET
- Letter of Recommendation receive from Production Engineer | Afaq Steel by Premium Batala
- Seminar on Additive Manufacturing of Bio-Materials
- Seminar on Business Intelligence & Optimization Solutions

LANGUAGES

- English
- Urdu
- Punjabi

REFERENCE

Dr. Ameer Farooq Assistant Professor & In-charge of Corrosion Control Research Cell (CCRC)
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